

SUMMARY

Procedural FX Artist with a technical background in Industrial Design. Specialising in Houdini-based simulations including Pyro, Vellum, and procedural workflows using VEX. Experienced in procedural problem-solving, complex geometry handling, and simulation optimisation through a multidisciplinary background in industrial design and prototyping. Focused on building efficient procedural systems and cinematic look development using Solaris, Karma and Redshift.

PROJECTS

Crowd Simulation & Procedural Animation

- Developed custom agent behavior logic using VEX for complex combat and movement patterns.
- Optimised large-scale simulations by implementing efficient caching and geometry proxy workflows.
- Created procedural transitions between animation states to ensure seamless visual storytelling.

Short Film – “UNEMPLOYED” (Practical FX, Look Development & Compositing)

- Produced ghost crowd effects and distorted motion sequences through practical cinematography and compositing workflows.
- Integrated procedural and composited visual elements into live-action footage using Houdini and Nuke.
- Researched cinematic look development workflows focused on visual consistency, atmosphere, and compositing.

Procedural FX Research & Personal Projects (2025. 01 – Present)

- Developed a Real-time Blob Tracking system using TouchDesigner for interactive visual installations.
- Researching Procedural HDA tools in Houdini to streamline repetitive FX tasks and asset generation.
- Exploring advanced simulation solvers and optimization techniques to enhance technical proficiency in Houdini.

SKILLS

- Houdini
 - SOPs, DOPs, VEX, Pyro, Vellum, RBD, Crowds, Solaris
- Rendering
 - Karma, Redshift, Arnold
- Compositing
 - Nuke, DaVinci Resolve, PFTrack, Adobe Premiere Pro
- 3D & Technical
 - Maya, Rhino, TouchDesigner, nTopology
- Programming / Workflow
 - Procedural modelling, Simulation Optimisation, Look Development

PROFESSIONAL EXPERIENCE

Industrial and CAD Designer

L. Corporation | Seoul, South Korea | 2020-2022

- Optimised high-resolution CAD data for technical visualisation and prototyping workflows.
- Developed complex geometry and production-ready 3D assets using nTopology and CAD tools.
- Collaborated with R&D teams on technical problem-solving, fabrication, and rapid prototyping.

Industrial Designer & Co-founder

Titan Corporation | Seoul, South Korea | 2019-2020

- Led full-cycle 3D asset production, from concept development to high-fidelity surface modelling and technical prototyping.
- Optimised 3D assets for manufacturing feasibility, managing complex surface modelling and assembly tolerances.
- Facilitated technical communication with manufacturers to ensure material integrity and efficient production workflows.

EDUCATION

Bournemouth University

Master of Arts in Digital Effects with Merit
2023 – 2024

Academic Credit Bank System (South Korea) National Institute for Lifelong Education (NILE)

Bachelor of Business Administration (G.P.A: 3.11/4.5)
2012 – 2016

PORTFOLIO LINKS

- [Portfolio Website](#)
- [VFX SHOWREEL](#)
- [CROWD SIMULATION SHOWREEL](#)
- [LINKEDIN](#)